Positivity in Algebraic Geometry I  Section 4.1 Reading Note:

Notation:

As always, we work with algebraic varieties defined over the complex numbers. Unless otherwise stated, all varieties are assumed to be non-singular.

## 4.1. Normal Crossing Divisors and resolution of Singularities.

**Definition 4.1.1. (Normal crossing divisor).** An effective divisor $D = \sum D_i$ on $X$ has *simple normal crossings* (and $D$ is a *simple normal crossing* or *SNC divisor*) if $D$ is reduced, each component $D_i$ is smooth, and $D$ is defined in a neighborhood of any point by an equation in local analytic coordinates of the type

$$z_1 \cdot \ldots \cdot z_k \;=\; 0$$

for some $k \leq n$. A divisor $E = \sum a_i D_i$ has *simple normal crossing support* if the underlying reduced divisor $\sum D_i$ has simple normal crossings. □

More general & more algebraic definition:

**Definition 0BI9.** Let $X$ be a locally Noetherian scheme. A *strict normal crossings divisor* on $X$ is an effective Cartier divisor $D \subset X$ such that for every $p \in D$ the local ring $\mathcal{O}_{X,p}$ is regular and there exists a regular system of parameters $x_1, \ldots, x_d \in \mathfrak{m}_p$ and $1 \leq r \leq d$ such that $D$ is cut out by $x_1 \ldots x_r$ in $\mathcal{O}_{X,p}$.

The two definitions coincide when $X$ in second one smooth variety over $k$.

Hironaka **Theorem 4.1.3. (Resolution of singularities).** *Let $X$ be an irreducible complex algebraic variety (possibly singular), and let $D \subseteq X$ be an effective Cartier divisor on $X$.*

(i). *There is a projective birational morphism*

$$\mu : X' \longrightarrow X,$$

a closed set of $X'$ hence as a closed subscheme (reduced)

*where $X'$ is non-singular and $\mu$ has divisorial exceptional locus* $\operatorname{except}(\mu)$, *such that*

$$\mu^* D + \operatorname{except}(\mu)$$

as a divisor.

*is a divisor with SNC support.*

(ii). *One can construct $X'$ via a sequence of blowings-up along smooth centers supported in the singular loci of $D$ and $X$. In particular,* ☆ stronger *one can assume that $\mu$ is an isomorphism over*

$$X \;-\; \big(\operatorname{Sing}(X) \cup \operatorname{Sing}(D)\big).$$

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statement in Birkar's MMP lecture note, only mentioned (ii) above.

**Theorem 1.9** (Hironaka resolution of singularities). *Assume that $X$ is a variety and $B$ is a divisor on $X$. Then there exists a projective birational morphism $\phi\colon W \to X$ such that $W$ is smooth and $\phi^{-1}\operatorname{Supp} B \cup \operatorname{Exc}(\phi)$ is a simple normal crossing divisor.*

In addition:

**Definition 6.1** (Pairs). A *pair* $(X, B)$ consists of a normal variety $X$ and a divisor $B = \sum b_i B_i$, $b_i \in [0,1]$, such that $K_X + B$ is $\mathbb{Q}$-Cartier, i.e., $m(K_X + B)$ is Cartier for some $m \in \mathbb{N}$.

**Definition 6.2** (Log resolution). A *log resolution of singularities* of a pair $(X, B)$ is a projective birational morphism $\phi\colon W \to X$ such that $W$ is smooth and $\phi^{-1}\operatorname{Supp} B \cup \operatorname{Exc}(\phi)$ is a simple normal crossing divisor.

Note that here $\operatorname{Exc}(\phi) = \bigcup C$, where $C \subseteq W$ are curves contracted by $\phi$. Log resolutions always exist by Hironaka (recall that we work over a field $k$ of characteristic 0).

**Corollary 4.1.4.** *Assume that $X$ is smooth, and that $\mu : X' \longrightarrow X$ is an embedded resolution of an effective divisor $D \subseteq X$ constructed as in 4.1.3 (ii) by a sequence of blowings-up along smooth centers.*

(i). *Denoting by $K_X$ and $K_{X'}$ the canonical divisors of $X$ and $X'$ respectively, one has*

$$\mu_* \mathcal{O}_{X'}(K_{X'}) \;=\; \mathcal{O}_X(K_X) \quad \text{and} \quad R^j \mu_* \mathcal{O}_{X'}(K_{X'}) \;=\; 0 \quad \text{for} \;\; j > 0.$$

(ii). *Assume that $X$ is projective and let $H$ be an ample divisor on $X$. Then for suitable integers $p \gg 0$ and $b_j \geq 0$,*

$$\mu^*(pH) \;-\; \sum b_j E_j$$

*is an ample divisor on $X'$, the sum being taken over the exceptional divisors of $\mu$.*

pf

Write $X' = X_n \xrightarrow{\pi_n} X_{n-1} \xrightarrow{\pi_{n-1}} X_{n-2} \xrightarrow{\pi_{n-2}} \cdots \xrightarrow{\pi_2} X_2 \xrightarrow{\pi_2} X_1 \xrightarrow{\pi_1} X_0 = X$

$\pi_j : X_j \longrightarrow X_{j-1}$ blowup of a smooth center $Z$ in $X_{j-1}$

i.e. $X_j \simeq \mathrm{Bl}_Z(X_{j-1}) \longrightarrow X_{j-1}$ with $Z$ smooth & $X_{j-1}$ smooth on every $x \in Z$

(in particular: $Z \hookrightarrow X_{j-1}$ is regular embedding)

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hope to prove $\begin{cases} \pi_*\mathcal{O}_{X_j}(KX_j) = \mathcal{O}_{X_{j-1}}(KX_{j-1}) \\ R^i\pi_*\mathcal{O}_{X_j}(KX_j) = 0 \quad \text{for } i>0. \end{cases}$

($\because \mathcal{O}_{Z,z} \cong \mathcal{O}_{X,z}/(f_{z,1})$ regular local ring; $f_{z,1}$ can be gen. by regular seq.)

we replace $X_j, X_{j-1}$ by $X', X$ for the time. $E$: exceptional div.

Claim: For $0 \le m \le c-1$ where $c = \operatorname{codim}_X Z$.

$\pi_*\mathcal{O}_{X'}(mE) = \mathcal{O}_X$. $\quad R^i\pi_*\mathcal{O}_{X'}(mE) = 0 \ \forall i \ge 1$

reason: $m=0$. by $X$ normal $X' \to X$ proj. birational. by Stein factorization

$\pi_*\mathcal{O}_{X'} = \mathcal{O}_X$.

$\forall x \in X\setminus Z$. $\because \pi$ restricted on $X'\setminus E \xrightarrow{\cong} X\setminus Z$ isom.

$\therefore (R^i\pi_*\mathcal{O}_{X'})_x = 0$.

$\forall z \in Z$. By formal functions Theorem

**28.3.2. Theorem on Formal Functions.** — *Suppose $\pi\colon X \to Y$ is a proper morphism of locally Noetherian schemes, $\mathcal{F}$ is a coherent sheaf on X, and $q \in Y$. Then (28.3.1.2) is an isomorphism for all $i \ge 0$.*

$$\left((R^i\pi_*\mathcal{O}_X)_{\hat{}}\right)^{\wedge} \cong \varprojlim_n H^i(X'_n, \mathcal{O}_{X'_n}) \qquad X'_n = X' \times_X \mathcal{O}_z/\mathfrak{m}_z^n.$$

Recall: $E = \operatorname{Proj}_Z(\mathcal{O}_X/\mathcal{I}_z \oplus \mathcal{I}_z/\mathcal{I}_z^2 \oplus \cdots) \to Z$.

Moreover, we see that the exceptional divisor can be described beautifully as a *Proj* over X:

(22.3.2.1)

$$\mathrm{E}_X\mathrm{Y} = \mathcal{P}roj_X\left(\mathcal{O}_Y/\mathcal{I} \oplus \mathcal{I}/\mathcal{I}^2 \oplus \mathcal{I}^2/\mathcal{I}^3 \oplus \cdots\right) = \mathcal{P}roj_X\left(\mathcal{O}_Y(\mathcal{I})_\bullet/(\mathcal{I}\mathcal{O}_Y(\mathcal{I})_\bullet)\right).$$

We will later see that in good circumstances (if X is a regular embedding in Y), this is a projectivization of a vector bundle (the "projectivized normal bundle"), see Exercise 22.3.D(a).

$\therefore X'_n \cong \mathbb{P}^{c-1}_{\mathcal{O}_z/\mathfrak{m}_z^n}$ $\quad H^i(X'_n, \mathcal{O}_{X'_n}) = 0 \ \forall n, i \ge 1$

$\therefore \left(R^i\pi_*\mathcal{O}_{X'}\right)_z = 0 \ \forall z \in Z$. As above $R^i\pi_*\mathcal{O}_{X'} = 0 \ \forall i \ge 1$

For $m \ge 1$. $\quad 0 \to \mathcal{O}_{X'}((m-1)E) \to \mathcal{O}_{X'}(mE) \to \mathcal{O}_E(mE) \to 0$.

first discuss as above: $\forall i \ge 0$. by Formal Function Theorem $\forall z \in Z$

$$\left(R^i\pi_*\mathcal{O}_{mE}(mE)_z\right)^{\wedge} \cong \varprojlim_n H^i(X'_n, \mathcal{O}_E(mE)|_{X'_n}). \quad X'_n \text{ as above}$$

(Rmk: $(R^i\pi_*\mathcal{O}_{mE}(mE))_x = 0$ for $x \notin Z$ because it supports at most on $Z$)

$$\cong \varprojlim_n H^i(\mathbb{P}^{c-1}_{\mathcal{O}_z/\mathfrak{m}_z^n}, \mathcal{O}_{\mathbb{P}}(-m))$$

$$= 0 \quad \text{for } m \in [0, c-1]$$

$0 \to \pi_*\mathcal{O}_{X'}((m-1)E) \to \pi_*\mathcal{O}_{X'}(mE) \to R^0\pi_*\mathcal{O}_E(mE) = 0$ (for $m \le c-1$)

$\to R^1\pi_*\mathcal{O}_{X'}((m-1)E) \to R^1\pi_*\mathcal{O}_{X'}(mE) \to R^1\pi_*\mathcal{O}_E(mE) = 0$

$\to \cdots\cdots$ do induction

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well known that $K_{X'} = \pi^* K_X + (c-1)E$.

$$\therefore R^i\pi_*\mathcal{O}_{X'}(K_{X'}) = R^i\pi_*\left(\mathcal{O}_{X'}((c-1)E)\otimes\pi^*\mathcal{O}_X(K_X)\right)$$
$$\overset{\text{proj}}{=} R^i\pi_*\mathcal{O}_{X'}((c-1)E)\otimes\mathcal{O}_X(K_X)$$
$$= \begin{cases} 0 & i\geq 1 \\ \mathcal{O}_X(K_X) & i=0. \end{cases}$$

For simplicity consider $X_2 \xrightarrow{\pi_2} X_1 \xrightarrow{\pi_1} X_0 = X$, with $\mu = \pi_1\circ\pi_2$.

**Lemma 01F6** (Relative Leray spectral sequence). *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of ringed spaces. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. There is a spectral sequence with*

$$E_2^{p,q} = R^p g_*(R^q f_*\mathcal{F})$$

*converging to $R^{p+q}(g\circ f)_*\mathcal{F}$. This spectral sequence is functorial in $\mathcal{F}$, and there is a version for bounded below complexes of $\mathcal{O}_X$-modules.*

$$E_2^{p,q} = \begin{cases} R^p\mu_*\mathcal{O}_{X_2}(K_{X_2}) = \mathcal{O}_X(K_X) & p=q=0 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore E_\infty^{p,q} = 0$ by definition of $E_r^{p,q}$ & get $E_r^{p,q} = 0$ $\forall (p,q)\neq(0,0)$, $r\geq 2$.

$\Rightarrow R^{p+q}\mu_*\mathcal{O}_{X_2}(K_{X_2}) = 0$ $\forall p+q\geq 1$.

(2) got proved ▨

only to prove: $X' = \mathrm{Bl}_Z X \xrightarrow{\pi} X$ with $Z\hookrightarrow X$ regular embedding.

for $p\gg 0$, $\pi^*pt - E$ is ample on $X'$.

reason: well-known that

The preimage of X isn't just any effective Cartier divisor; it corresponds to the invertible sheaf $\mathscr{O}(1)$ on this *Proj*. Indeed, $\mathscr{O}(1)$ corresponds to taking our graded ring, chopping off the bottom piece, and sliding all the graded pieces to the left by 1 (§15.1.4); it is the invertible sheaf corresponding to the graded module

$$\mathrm{IB}(\mathrm{I})_\bullet = \mathrm{I}\oplus\mathrm{I}^2\oplus\mathrm{I}^3\oplus\cdots$$

(where that first summand I has grading 0). But this can be interpreted as the scheme-theoretic preimage of X, which corresponds to the ideal I of B:

$$\mathrm{IB}(\mathrm{I})_\bullet = \mathrm{I}\left(\mathrm{B}\oplus\mathrm{I}\oplus\mathrm{I}^2\oplus\cdots\right)\hookrightarrow \mathrm{B}\oplus\mathrm{I}\oplus\mathrm{I}^2\oplus\cdots = \mathrm{B}(\mathrm{I})_\bullet.$$

Thus the scheme-theoretic preimage of $\mathrm{X}\hookrightarrow\mathrm{Y}$ to

$$\mathscr{P}roj(\mathscr{O}_\mathrm{Y}(\mathscr{I})_\bullet) = \mathscr{P}roj(\mathscr{O}_\mathrm{Y}\oplus\mathscr{I}\oplus\mathscr{I}^2\oplus\cdots),$$

corresponding to the ideal sheaf corresponding to $\mathscr{I}\mathscr{O}_\mathrm{Y}(\mathscr{I})_\bullet = \mathscr{I}\oplus\mathscr{I}^2\oplus\mathscr{I}^3\oplus\cdots$, is an effective Cartier divisor in class $\mathscr{O}(1)$. Once we have verified that this construction indeed yields the blow-up, this divisor will be our exceptional divisor $\mathrm{E_X Y}$.

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1. $\mathcal{O}_X(-b) = \mathcal{I}_b = (\mathcal{O}_X(1))$ is b-ample.

**Proposition 13.65.** *Let $S$ be a quasi-compact scheme, let $f\colon X \to Y$ and $g\colon Y \to S$ be morphisms of finite type. Let $\mathscr{L}$ be an $f$-ample $\mathcal{O}_X$-module and let $\mathscr{K}$ be a $g$-ample $\mathcal{O}_Y$-module. Then there exists an integer $n_0$ such that $\mathscr{L} \otimes f^*(\mathscr{K})^{\otimes n}$ is ample for $g \circ f$ for all $n \geq n_0$.*

(Ref: Conrad I). (?!) got proved ▨

**Proposition 1.7.10.** *Consider a morphism*

$$f : X \longrightarrow T$$

*of projective schemes. Let $L$ be a line bundle on $X$, and let $A$ be an ample line bundle on $T$. Then $L$ is $f$-ample if and only if $L \otimes f^*\left(A^{\otimes m}\right)$ is an ample line bundle on $X$ for all $m \gg 0$.*

(Ref: Lazersfeld I).

4.1.B Covering Lemmas.

(previous: important tool in proving Vanishing Theorems)

**Proposition 4.1.6. (Cyclic coverings).** *Let $X$ be a variety, and $L$ a line bundle on $X$. Suppose given a positive integer $m \geq 1$ plus a non-zero section*

$$s \in \Gamma\left(X, L^{\otimes m}\right)$$

*defining a divisor $D \subseteq X$. Then there exists a finite flat covering*

$$\pi : Y \longrightarrow X,$$

*where $Y$ is a scheme having the property that the pullback $L' = \pi^* L$ of $L$ carries a section*

$$s' \in \Gamma\left(Y, L'\right) \quad \text{with} \quad \left(s'\right)^m = \pi^* s.$$

*The divisor $D' = \operatorname{div}(s')$ maps isomorphically to $D$. Moreover if $X$ and $D$ are non-singular, then so too are $Y$ and $D'$.*

construction: consider $\underline{\mathrm{Spec}}_{\mathcal{O}_X} \mathrm{Sym}\mathcal{L}^\vee \xrightarrow{p} X$.

locally: $\{U_i = \mathrm{Spec}A_i\}$ open covering of $X$, & $\mathcal{L}|_{U_i} = \widetilde{A_i e_i}$

$p^{-1}(U_i) \to U_i$ is just $\mathrm{Spec}A_i[e_i^*] \to \mathrm{Spec}A_i$

$p^*\mathcal{L}\big|_{p^{-1}(U_i)} = \widetilde{A_i[e_i^*] \cdot e_i}$ $\{e_i^* \cdot e_i\}$ can be glued together

hence form a "tautological" section $T \in \Gamma(\underline{\mathrm{Spec}}_{\mathcal{O}_X} \mathrm{Sym}\mathcal{L}^\vee, p^*\mathcal{L})$

consider $T^m - p^*s \in \Gamma(\underline{\mathrm{Spec}}_{\mathcal{O}_X} \mathrm{Sym}\mathcal{L}^\vee, p^*\mathcal{L}^{\otimes m})$

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& $Y = \mathrm{div}(T^m - p^*s)$

locally: $Y \cap p^{-1}(U_i) = Y \times_X U_i = \mathrm{Spec}\, A_i[e_i^*]/(e_i^{*m} - a_i)$ where $s \in \Gamma(X, \mathcal{L}^{\otimes m})$, $s|_{U_i} = a_i \cdot e_i^m$.

$A_i[e_i^*]/(e_i^{*m} - a_i)$ is finite over $A_i$ & as a free $A_i$-module $\forall i$

hence $Y \xrightarrow{\pi} X$ is finite flat surjective ——— ①

$Y \hookrightarrow \mathrm{Spec}_{\mathcal{O}_X} \mathrm{Sym}\, \mathcal{L}^\vee \xrightarrow{p} X$. $s' = T|_Y$ then $s'^m = T^m|_Y = p^*s|_Y = \pi^*s$. ——— ②

locally: $s'|_{\pi^{-1}U_i} = \overline{e_i^*} \cdot e_i \in A_i[e_i^*]/(e_i^{*m} - a_i) \cdot s_i$

so $\mathrm{div}\, s'|_{\pi^{-1}(U_i)} = \mathrm{Spec}\, A_i[e_i^*]/(e_i^{*m} - a_i, e_i^*) = \mathrm{Spec}\, A_i/(a_i) = \mathrm{div}\, s|_{U_i}$.

hence $\mathrm{div}\, s' \cong \mathrm{div}\, s$ as schemes ——— ③

Moreover: If $X$ and $D$ are nonsingular when $D = \mathrm{div}\, s$. $D' = \mathrm{div}\, s'$. $D'$ nonsing. by ③

Now prove $Y$ is nonsingular:

Case 1: $x \notin D$. Choose an aff. neighborhood $U = \mathrm{Spec}\, A$ of $x$ in $X \setminus D$

$\pi^{-1}(U) \longrightarrow U$ is actually $\mathrm{Spec}\, A[e^*]/(e^{*m} - a) \longrightarrow \mathrm{Spec}\, A$

with $\mathcal{L}|_U = \widetilde{Ae}$. $s|_U = ae^m$ $\forall U \subseteq X \setminus D$. hence $a \in A^*$

$\Omega_{Y/X}|_{\pi^{-1}(U)} = \Omega_{\pi^{-1}(U)/U} \cong \left(A[e^*]/(e^{*m} - a \cdot me^{*m-1})\right)^{\sim} \xrightarrow{\varphi} 0$

(need $a \in A^*$ then $\mathbb{Q} \neq 0$)

**10.3.** A morphism $f: X \to Y$ of schemes of finite type over $k$ is *étale* if it is smooth of relative dimension 0. It is *unramified* if for every $x \in X$, letting $y = f(x)$, we have $\mathfrak{m}_y \cdot \mathcal{O}_x = \mathfrak{m}_x$, and $k(x)$ is a separable algebraic extension of $k(y)$. Show that the following conditions are equivalent:

(i) $f$ is étale;
(ii) $f$ is flat, and $\Omega_{X/Y} = 0$;
(iii) $f$ is flat and unramified.

(Reference: GTM52 III)

$\therefore \pi|_{\pi^{-1}(X \setminus D)}: \pi^{-1}(X \setminus D) \longrightarrow X \setminus D$ is étale.

hence by $X$ nonsingular. $\pi^{-1}(X \setminus D)$ is nonsingular.

Case 2: $x \in D$. $s_x = f \cdot e_x \in \mathcal{O}_{X,x} \cdot e_x = \mathcal{L}_x$. $f \in \mathfrak{m}_x$.

$D$ nonsingular $\Rightarrow$ $\mathcal{O}_{X,x}/(f)$ is regular local ring

by $\mathcal{O}_{X,x}$ regular local ring. Fact: $f \in \mathfrak{m}_x \setminus \mathfrak{m}_x^2$.

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$(f, f_2, f_3, \cdots f_n)$ form a regular seq. of $\mathcal{O}_{X,x}$.

locally: a prime ideal $q$ of $A[e^*]/(e^{*m}-f)$ whose intersection with $A$ is a prime ideal $p$ containing $f$ must be $(e^*, p)$

$(f \in q \Rightarrow e^{*m} \in q \Rightarrow e^* \in q)$

$\therefore$ there's only one $y \in Y$ maps to $x \in X$ by $\pi$ and $m_y = (e^*, \overline{t_y}f_2, \cdots \overline{t_y}f_n)$

it is a regular sequence of $\mathcal{O}_{Y,y}$

$\therefore$ $Y$ is nonsingular at $y \in \pi^{-1}(D)$. ▨

Rmk: · $\pi^* D = m\mathcal{D}$ while $D \cong D'$ as scheme.

· In fact $Y$ is a variety under "Moreover" condition.

fin. type: $Y \xrightarrow{\text{fin.}} X \xrightarrow{\text{fin. type}} \mathbb{C}$, reduced: by nonsingular., sep: $Y \to X \xrightarrow{\text{sep.}} \mathbb{C}$ (fin. ⇒ proper ⇒ sep.)

· $Y$ is integral if $X$ integral + "Moreover" condition.

$\forall x \in D (\neq \emptyset)$ $f \in m_x \setminus m_x^2$.

by Eisenstein $e^{*m} - f$ is irreducible in $(\mathrm{Frac}\,\mathcal{O}_{X,x})[e^*]$ (= $K(X)[e^*]$)

$\eta$: generic pt of $X$. $Y_\eta \cong \mathrm{Spec}\, K(X)[e^*]/(e^{*m} - f)$ is integral

**Proposition 3.8.** *Let $Y$ be a scheme having only a finite number of irreducible components. Let $f : X \to Y$ be a flat morphism. Let us suppose that $Y$ is reduced (resp. irreducible; resp. integral), and that the generic fibers of $f$ are also reduced (resp. irreducible; resp. integral); then $X$ is reduced (resp. irreducible; resp. integral).*

(Ref: LiuQing Chapter 4) immediately have $Y$ is integral.

**Remark 4.1.8.** It will be useful to note that normal crossing divisors pull back nicely under the covering just constructed. Specifically, assume in the situation of 4.1.6 that $X$ and $D$ are smooth, and suppose that $\sum D_i$ is a reduced effective divisor on $X$ such that $D + \sum D_i$ has simple normal crossings. Put $D'_i = \pi^* D_i$. Then it follows from the local discussion that $D' + \sum D'_i$ is a simple normal crossing divisor on $Y$. □

pf $D_0 = D$

$\forall x \in D_0 \cup D_1 \cup \cdots \cup D_q$ with $D_i$ induces $f_i \in m_x$ $f_0 = f$

$\{f_0, f_1 \cdots f_q\}$ form a subset of a regular sequence, enlarge it to $\{f_0 \cdots f_n\}$ (after removing $\mathcal{O}_{X,x}^*$, omitted here and from now on).

If $x \in D$, then $\{e^*, \overline{t_y}f_1 \cdots \overline{t_y}f_n\}$ form regular sequence at the only $y \in Y$ mapping to $x$ by $\pi$

$y \in D'$.

If $x \notin D$. $\Rightarrow \pi$ is étale on $\pi^{-1}(X \setminus D)$. ($f$ is removed).

$\forall y \in Y$. mapping to $x$. $(\pi_y^\# f_1, \dots, \pi_y^\# f_n)$ forms a regular seq. of $m_y$

($(\pi_y^\# f_1, \dots, \pi_y^\# f_n) = m_y$ by unramified
& regular seq. by flatness
$a \in A$ s.t $A/I \xrightarrow{\cdot a} A/I$ isom (i.e. $a$ is a unit in $A/I$)
(then if $A \to B$ flat. then $B/IB \xrightarrow{\cdot \varphi(a)} B/IB$ isom. i.e. $\varphi(a)$ is unit in $B/IB$))

$\therefore D' + \sum \pi^* D_i$ SNC. ▨

**Lemma 4.1.14. (Injectivity lemma).** *Let $f : Y \longrightarrow X$ be a finite surjective morphism of irreducible complex projective varieties, with $X$ normal, and let $E$ be a vector bundle on $X$. Then the natural homomorphism*

$$H^j\big(X, E\big) \longrightarrow H^j\big(Y, f^* E\big)$$

*induced by $f$ is injective. In particular, if*

$$H^j\big(Y, f^* E\big) \;=\; 0 \quad \text{for some } j \geq 0,$$

*then $H^j\big(X, E\big) = 0$.*

pf. Try to construct $\mathcal{O}_X \longrightarrow f_*\mathcal{O}_Y \xrightarrow{Tr} \mathcal{O}_X$

$\{U_i = \mathrm{Spec} A_i\}$ covering of $X$. $f^{-1}(U_i) = \mathrm{Spec} B_i$ $B_i$ is fin. extension of $A$ by $f$ fin surj.

consider $\Gamma(U_i, f_*\mathcal{O}_Y) = B \xrightarrow{Tr(U_i)} \Gamma(U_i, \mathcal{O}_X) = A_i$.

$b \longmapsto Tr_{L/K(X)}(b) = b_1 + b_2 + \dots + b_m$ $L :=$ Galois closure of $K(Y)/K(X)$

$b$ is fin. over $A$. $\therefore b_i$ are fin. over $A$. $\therefore Tr(U_i)(b)$ fin. over $A$.

by $A$ normal. $\therefore Tr(U_i)(b) \in A$. $Tr(U_i)$ is well-defined.

Because all elements in $B_i$ can be seen as an element in $K(Y)$

$\therefore Tr(U_i)$ can be glued together, $\therefore Tr$ is well-defined.

$\mathcal{O}_X \to f_*\mathcal{O}_Y \to \mathcal{O}_X$ is actually $\mathcal{O}_X \xrightarrow{\cdot [L:K(X)]} \mathcal{O}_X$

So $\mathcal{O}_X \to f_*\mathcal{O}_Y \xrightarrow{\cdot \frac{1}{[L:K(X)]} Tr} \mathcal{O}_X$ (id)

$\otimes \mathcal{E} \Rightarrow \mathcal{E} \to f_*\mathcal{O}_Y \otimes \mathcal{E} \cong f_* f^* \mathcal{E} \xrightarrow{\cdot \frac{1}{n} Tr \otimes id} \mathcal{E}$ (id)

$\therefore \mathcal{E}$ is a summand of $f_* f^* \mathcal{E}$. $\therefore H^i(X, \mathcal{E}) \hookrightarrow H^i(X, f_* f^* \mathcal{E}) = H^i(Y, f^* \mathcal{E})$.

▨

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**Theorem 4.2.1. (Kodaira vanishing theorem).** *Let $X$ be a smooth irreducible complex projective variety of dimension $n$, and let $A$ be an ample divisor on $X$. Then*

$$H^i\big(X, \mathcal{O}_X(K_X + A)\big) \;=\; 0 \quad \text{for} \;\; i > 0.$$

*Equivalently,*

$$H^j\big(X, \mathcal{O}_X(-A)\big) \;=\; 0 \quad \text{for} \;\; j < n = \dim X.$$

Rmk: only discuss how to reduce the prof here, the residue part was mentioned in previous seminars.

Choose $m \gg 0$ s.t. $mA$ very ample

by Bertini, we can choose $s \in \Gamma(X, \mathcal{O}_X(mA))$ s.t. $\operatorname{div} s$ smooth, denoted by $D$

by the construction of cyclic covering, $\exists\, Y \xrightarrow{\pi} X$ finite flat surj. $Y$ is smooth ver. (by the Rmk given after the prof)

recall that we can find $s' \in \Gamma(Y, \pi^*\mathcal{O}_X(A))$ s.t. $s'^m = \pi^* s \in \Gamma(Y, \pi^*\mathcal{O}_X(mA))$

$\operatorname{div} s' = D'$. $\therefore D' \sim \pi^* A$.

If we proved $H^i(Y, \mathcal{O}_Y(-D')) = 0$ for $i < \dim Y = \dim X$ (∵ $\pi$ finite surj.)

$\mathcal{O}_Y(-D') = \pi^*\mathcal{O}_X(-A)$

by projection formula $H^i(X, \mathcal{O}_X(-A)) = 0$.

(can be used by $X$ sm. hence normal.)

By $D' \cong D$ as scheme with $D$ smooth $\therefore$ WMA $A$ sm. effective.

residue prove based on Complex Geometry facts ..... ▨

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**Theorem 4.1.10. (Bloch–Gieseker coverings).** *Let $X$ be an irreducible quasi-projective variety of dimension $n$, let $L$ be a line bundle on $X$, and fix any positive integer $m$. Then there exists a reduced and irreducible variety $Y$, a finite flat surjective mapping*

$$f : Y \longrightarrow X,$$

*together with a line bundle $N$ on $Y$ such that*

$$f^*L \;=\; N^{\otimes m}.$$

*If $X$ is non-singular then we can take $Y$ to be so as well. In this case, given a simple normal crossing divisor*

$$D \;=\; \sum D_i$$

*on $X$, we can arrange that its pullback $f^*D$ be a simple normal crossing divisor on $Y$. Finally, if $\dim X \geq 2$ then we can further ensure that the preimage $D'_i = f^*D_i$ of each irreducible component of $D$ is irreducible.*

Step 1: First prove for $L$ very ample.

consider $Y_g \xrightarrow{\phi_g} \mathbb{P}^r$, $g \in GL_{r+1}$

$f_g \downarrow \square \downarrow \nu$, $\nu : [a_0 : \cdots : a_r] \mapsto [a_0^m : a_1^m : \cdots : a_r^m]$ Veronese morphism

$X \hookrightarrow \mathbb{P}^r$, $g$

$\mathcal{L} = \phi^* \mathcal{O}_{\mathbb{P}^r}(1)$

consider $N = \phi_g^* \mathcal{O}_{\mathbb{P}^r}(1)$. then $N^{\otimes m} = \phi_g^* \mathcal{O}_{\mathbb{P}^r}(m) = \phi_g^* \nu^* g^* \mathcal{O}_{\mathbb{P}^r}(1) = f_g^* \phi^* \mathcal{O}_{\mathbb{P}^r}(1) = f_g^* \mathcal{L}$.

$\nu_g := g \circ \nu$ is finite flat surj. by BC. $f_g$ is also fin. flat surj.

(need the base field closed (C))

Step 1.1 Goal: Choose $g$ s.t $Y_g$ above is irreducible & reduced.

different description of $Y_g$: (geometrically thinking).

from above Cartesian diagram, elements are of the form:

$$\{(x,y) \mid \phi(x) = \nu_g(y),\ x \in X,\ y \in \mathbb{P}^n\}$$
$$\cong \{((x,z),w) \mid x \in X,\ z \in \mathbb{P}^n,\ w \in \mathbb{P}^r,\ \phi(x) = w,\ \nu(z) = g^{-1}(w)\}$$

$(x,y) \mapsto (x, y), \phi(x))$

$(x,z) \mapsto (x,z), w)$

∴ we have the following Cartesian diagram:




[Conrad]. use for $X \subseteq \mathbb{P}^r$ here.

**Proposition 5.51.** *Let $X$ be a $k$-scheme. Then the following assertions are equivalent.*

(i) *$X$ is geometrically integral.*
(ii) *For every integral $k$-scheme $Y$ the product $X \times_k Y$ is integral.*
(iii) *$X$ is integral and if $\eta$ is the generic point of $X$, then $k$ is algebraically closed in $\kappa(\eta)$ and $\kappa(\eta)$ is separable over $k$.*
(iv) *There exists an algebraically closed extension $\Omega$ of $k$ such that $X_\Omega$ is integral.*
(v) *For every finite extension $K$ of $k$, $X_K$ is integral.*

Apply:

**Example 3.3.10. (Irreducibility of inverse image of general graph).** Let $X$ be an irreducible quasi-projective variety, and let

$$f : X \longrightarrow \mathbf{P}^r \times \mathbf{P}^r$$

be a morphism such that $\dim \overline{f(X)} > r$. Given $g \in G =_{\text{def}} \mathrm{GL}(r+1)$, which we view in the evident way as an automorphism of $\mathbf{P}^r$, let $\Gamma_g \subseteq \mathbf{P}^r \times \mathbf{P}^r$ be its graph. Then $f^{-1}(\Gamma_g)$ is irreducible for general $g \in G$. (Returning to the

∴ $Y_g$ is irreducible for general $g \in G(k)$

For reducedness:

Reference: Kleiman's The transversality of a general translate.

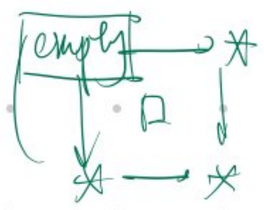
2. THEOREM: *Let $G$ be an integral algebraic group scheme, $X$ an integral algebraic scheme with a transitive $G$-action. Let $f: Y \to X$ and $g: Z \to X$ be two maps of integral algebraic schemes. For each rational point $s$ of $G$, let $sY$ denote $Y$ considered as an $X$-scheme via the map, $y \mapsto sf(y)$.*

(i) *Then, there exists a dense open subset $U$ of $G$ such that, for each rational point $s$ in $U$, either the fibered product, $(sY) \times_X Z$, is empty or it is equidimensional and its dimension is given by the formula,*

$$\dim((sY) \times_X Z) = \dim(Y) + \dim(Z) - \dim(X).$$

(ii) *Assume the characteristic is zero, and $Y$ and $Z$ are regular. Then, there exists a dense open subset $U$ of $G$ such that, for each rational point $s$ in $U$, the fibered product, $(sY) \times_X Z$, is regular.* (resp. reduced)

Rmk: (ii) is also important. It tells us that for general $g$, $Y_g$ is reasonable.




(worst happen)

using (ii), reducedness holds.

(ii) The reasonability of replacing "regular" by "reduced" is mentioned by Kleiman as following:

7. REMARK: It is easy to prove similarly the analogous forms of (1, (ii)), of (2, (ii)), of (4, (ii)), of (5) and a strengthened, analogous form of (6) with the adjective 'regular' replaced throughout by the adjective 'Cohen-Macaulay' (resp. 'normal', 'reduced', 'locally integral' etc.); in the new from of (6), the subscheme of $Z$ can be proved Cohen-Macaulay, etc., everywhere because the corresponding Schubert variety is so [3, 5, 7, 8, or 9]. In the forms with 'Cohen-Macaulay', the characteristic can be arbitrary.

Step 2 : $X$ regular. ($Y_g$ can be regular as above (iii))

After fixing SNC $D$ on $X$. Goal: general $g$, $f_g^* D$ is SNC on $Y_g$.

SNC has the following criterion.

**Lemma 0BIA.** *Let $X$ be a locally Noetherian scheme. Let $D \subset X$ be an effective Cartier divisor. Let $D_i \subset D, i \in I$ be its irreducible components viewed as reduced closed subschemes of $X$. The following are equivalent*

(1) *$D$ is a strict normal crossings divisor, and*

(2) *$D$ is reduced, each $D_i$ is an effective Cartier divisor, and for $J \subset I$ finite the scheme theoretic intersection $D_J = \bigcap_{j \in J} D_j$ is a regular scheme each of whose irreducible components has codimension $|J|$ in $X$.* or $D_J$ empty

$D = \sum_{i \in I} D_i$ consider

$$\begin{array}{ccccc} D_i \times_X Y_g & \hookrightarrow & Y_g & \xrightarrow{\varphi_g} & \mathbb{P}^r \\ \downarrow & \square & \downarrow f_g & \square & \downarrow \nu_g \\ D_i & \hookrightarrow & X & \xrightarrow{\varphi} & \mathbb{P}^r \end{array} \quad \rightsquigarrow (1)$$

$$\begin{array}{ccccccc} Z_g & \hookrightarrow & D_{i_1} \times_X Y_g \times_X \cdots \times_X D_{i_r} \times_X Y_g & \hookrightarrow & Y_g & \longrightarrow & \mathbb{P}^r \\ \downarrow & & \downarrow & \square & \downarrow & \square & \downarrow \nu_g \\ Z & \hookrightarrow & D_{i_1} \times_X \cdots \times_X D_{i_r} & \hookrightarrow & X & \longrightarrow & \mathbb{P}^r \end{array} \quad \rightsquigarrow (2)$$

$D$ SNC. by Lemma: $D$ reduced.

$\forall J \subseteq I$ $J = \{i_1, \ldots, i_r\}$. $D_{i_1} \times_X \cdots \times_X D_{i_r}$ regular with irreducible components codim $= |J|$

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By previous progress. from ① & ② for general $g \in G_r$. the following simultaneously holds:

- $D_i g$ reduced (& irr. when $\dim X \geq 2$ (i.e. $\dim D_i \geq 1$))
- $\forall J = \{i_1 \dots i_r\}$ $D_{i_1 g} \times_{Y_g} \dots \times_{Y_g} D_{i_r g}$ are regular & pure $\dim = \dim X - r$.

⇓

WARNING: can't use Kleiman on $D_{i_1} \times_X \dots \times_X D_{i_r}$ directly! (∵ $D_{i_1} \times_X \dots \times_X D_{i_r}$ not irr. hence not integral. can't apply Kleiman's Theorem.)

But this condition can still be satisfied. (one of the condition in Kleiman Thm.)

Moreover: $D_{i_1} \times_X \dots \times_X D_{i_r}$ regular & if not empty.

**5.4.B.** EXERCISE. Show that a Noetherian scheme is normal if and only if it is the finite disjoint union of integral Noetherian normal schemes. (Hint: Exercise 5.3.C.)

$\therefore D_{i_1} \times_X \dots \times_X D_{i_r} = \coprod_{j=1}^{n_J} Z_{J,j}$ each $Z_{J,j}$ irreducible regular var.

By Kleiman $Z_{J,j,g}$ regular (& irreducible if $\dim Z_{J,j} \geq 1$ i.e. $\dim X \geq 2$)

$\therefore D_{i_1 g} \times_X \dots \times_X D_{i_r g} = \coprod_{j=1}^{n_J} Z_{J,j,g}$ regular.

& $Z_{J,j,g}$ pure $\dim = \dim \mathbb{P}^r + \dim Z_{J,j} - \dim \mathbb{P}^r = n - |J|$ by Kleiman(3).

- $Z_{J,j,g}$ are all nonempty!

$\therefore$ In this case $f_g^* D_i = D_{i g}$ $f_g^* D = \sum f_g^* D_i = \sum D_{i g}$ satisfies Lemma condition (2).

hence $f_g^* D$ is SNC.

moreover $\dim X \geq 2$. then $f_g^* D_i = D_{i g}$ are irreducible reduced

Step 3. general $\mathscr{L}$. $\mathscr{L} = \mathscr{A} \otimes \mathscr{B}^\vee$ $\mathscr{A}, \mathscr{B}$ very ample

$$\begin{array}{ccc} Y_1 & \longrightarrow & \mathbb{P}^r \\ \downarrow f_1 & & \downarrow \nu_{g_1} \\ X & \xrightarrow{\phi_1} & \mathbb{P}^r \end{array}$$

s.t. $f_1^* \mathscr{B} = \mathscr{N}^{\otimes m}$ $f_1^* \mathscr{A}$ ample. by $f_1$ finite

$f_1^* D$ SNC. ($f_1^* D_i$ irr. if $\dim X \geq 2$)

$f_1^* \mathscr{L} = f_1^* \mathscr{A} \otimes \mathscr{N}^{\vee \otimes m} = \underbrace{(f_1^* \mathscr{A})^{\otimes (lm+1)}}_{\text{very ample}} \otimes \underbrace{(f_1^* \mathscr{A}^{\otimes l} \otimes \mathscr{N})^{\vee \otimes m}}_{\text{"}\mathscr{N}\text{"}}$

$$\rightsquigarrow \begin{array}{ccc} Y_2 & \longrightarrow & \mathbb{P}^s \\ \downarrow f_2 & \square & \downarrow \nu_{g_2} \\ Y_1 & \xrightarrow{\phi_2} & \mathbb{P}^s \end{array}$$

$f_2^* ((f_1^* \mathscr{A})^{\otimes (lm+1)}) = \mathscr{M}^{\otimes m}$

Choose $f = f_1 \circ f_2$. then $f^* \mathscr{L} = (\mathscr{M} \otimes f_2^* \mathscr{N})^{\vee \otimes m}$

$f^* D = f_2^* (f_1^* D)$ SNC ($f^* D = \sum f^* D_i = \sum \underbrace{f_2^* f_1^* D_i}$)

irr. when $\dim X \geq 2$.

□

**Proposition 4.1.12. (Kawamata coverings).** *Let $X$ be a non-singular quasi-projective variety, let $D = \sum_{i=1}^{t} D_i$ be a simple normal crossing divisor on $X$, and fix positive integers $m_1, \ldots, m_t > 0$. Then there exists a smooth variety $Y$ and a finite flat covering $h : Y \longrightarrow X$ with the property that*

$$h^* D_i \;=\; m_i D_i'$$

*for some smooth divisor $D_i'$ on $Y$, where $D' = \sum D_i'$ has simple normal crossings.*

pf.

- $X_1 \xrightarrow{f_1} X$ s.t. $\begin{cases} f_1^* \mathcal{O}_X(D_1) = \mathcal{N}_1^{\otimes m_1} \\ f_1^* D \text{ SNC.} \end{cases}$
- $X_2 \xrightarrow{f_2} X_1$ s.t. $\begin{cases} f_2^* f_1^* \mathcal{O}_X(D_1) = \mathcal{N}_2^{\otimes m_2} \\ f_2^* f_1^* D \text{ SNC} \end{cases}$

  ⋮
- $X_n \xrightarrow{f_n} X_{n-1} \cdots \xrightarrow{f_2} X_1 \xrightarrow{f_1} X$ (with $f = f_1 \circ \cdots \circ f_n$) s.t. $\begin{cases} f^* \mathcal{O}_X(D_i) = \mathcal{N}_i^{\otimes m_i} \\ f^* D \text{ SNC} \end{cases}$

Bloch–Gieseker Covering.

- $Y_1 \xrightarrow{g_1} X_n$ s.t. $f^* D_1 =: D_1'$ SNC.
  $\begin{cases} g_1^* D_1' = m_1 \widetilde{D_1'} & D_1' \cong \widetilde{D_1'} \text{ as scheme} \\ \widetilde{D_1'} + g_1^* f^* D_{i+1} \to g_1^* f^* D_n \text{ SNC.} \end{cases}$
- $Y_2 \xrightarrow{g_2} Y_1$ s.t. $g_1^* f^* D_2 = D_2'$. SNC
  $\begin{cases} g_2^* D_2' = m_2 \widetilde{D_2'} & D_2' \cong \widetilde{D_2'} \text{ as scheme} \\ g_2^* \widetilde{D_1'} + \widetilde{D_2'} + g_2^* g_1^* f^* D_{i+1} \to g_2^* g_1^* f^* D_n \text{ SNC.} \end{cases}$

  ⋮
- $Y_n \xrightarrow{g_n} Y_{n-1} \to \cdots \to Y_1 \xrightarrow{g_1} X_n \xrightarrow{f_n} X$ (with $g = g_1 \circ \cdots \circ g_n$)
  $\begin{cases} g^* D_i = m_i \widetilde{D_i} & (\widetilde{D_i} \neq \widetilde{D_i'}) \\ \sum \widetilde{D_i} \text{ SNC.} \end{cases}$ ∎

Cyclic Covering

Rmk: $X$ proj. (quasi-) each $X_i$ proj. (f-) $Y_1 \to X_n$ finite, $\Rightarrow Y_1$ proj. (f-) (fin. pullback of ample is ample)

Similarly $Y_2, Y_3 \cdots Y_n$ proj. (f-)

(Recall/Remind: pullback keeps SNC needs $Y_j, X_j$ all regular. It is proved before)

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**Lemma 4.3.5. (Norimatsu's lemma).** *Let $X$ be a smooth projective variety, $A$ an ample divisor on $X$, and $E$ an SNC divisor on $X$. Then*

$$H^i\big(X, \mathcal{O}_X(K_X + A + E)\big) \;=\; 0 \quad \text{for} \;\; i > 0,$$

*or equivalently,*

$$H^j\big(X, \mathcal{O}_X(-A - E)\big) \;=\; 0 \quad \text{for} \;\; j < \dim X.$$

*Proof.* Write $E = \sum_{i=1}^{t} E_i$. One argues by induction on the number $t$ of components of $E$, the case $t = 0$ being Kodaira vanishing. Assuming the result known for normal crossing divisors with $\leq k$ components, consider the natural exact sequence.

$$0 \longrightarrow \mathcal{O}_X\Big(-A-\sum_{i=1}^{k+1} E_i\Big)\longrightarrow\mathcal{O}_X\Big(-A-\sum_{i=1}^{k} E_i\Big) \longrightarrow \mathcal{O}_{E_{k+1}}\Big(-A-\sum_{i=1}^{k} E_i\Big) \longrightarrow 0.$$

The induction hypothesis, applied on $E_{k+1}$ and $X$, gives

$$H^j\Big(\, E_{k+1}\,,\, \mathcal{O}_{E_{k+1}}\big(\,-\,A\,-\sum_{i=1}^{k} E_i\big)\Big) \;=\; 0 \quad \text{for} \;\; j < (n-1),$$

$$H^j\Big(\, X\,,\, \mathcal{O}_X\big(\,-\,A\,-\sum_{i=1}^{k} E_i\big)\Big) \;=\; 0 \quad \text{for} \;\; j < n.$$

This yields the analogous vanishing for the term on the left, and the lemma follows. □

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**Theorem 4.3.1. (Vanishing for nef and big divisors).** *Let $X$ be a smooth complex projective variety of dimension $n$, and $D$ a nef and big divisor on $X$. Then*

$$H^i\big(X, \mathcal{O}_X(K_X + D)\big) \;=\; 0 \quad \text{for} \;\; i > 0. \tag{4.4}$$

*Equivalently,*

$$H^j\big(X, \mathcal{O}_X(-D)\big) \;=\; 0 \quad \text{for} \;\; j < n = \dim X. \tag{4.5}$$

pf. Since $D$ is big. $\exists m \in \mathbb{N}^*$ s.t. $mD \sim A + E$ w. $A$ ample, $E \geq 0$.

case 1 : if $E$ SNC support $E = \sum_{i=1}^{l} a_i E_i$

By Kawamata Covering. $\exists\, h : Y \longrightarrow X$ finite flat surj.

w/ $Y$ irreducible projective smooth var.

s.t. $h^* E_i = m \cdot \prod_{j \neq i} a_j \cdot E_i'$ $\quad \sum E_i'$ SNC

$\therefore mh^*D \sim h^*A + \big(m \prod_{i=1}^{l} a_i\big) \cdot \sum_{i=1}^{l} E_i'$

Write $D' = h^*D$. $E' = \sum_{i=1}^{l} E_i'$. $A' = h^*A$ $a = \prod_{i=1}^{l} a_i$.

we claim: $D' - E'$ is ample.

( reason: $ma(D' - E') \sim \underbrace{A' + (ma - m)D'}_{\text{ample + nef is ample}}$ )

So $H^j(Y, \mathcal{O}(-D')) = H^j(Y, \mathcal{O}(-(D' - E' - E'))) = 0 \quad j < \dim Y = \dim X$

by above Lemma.

By inj. lemma. $H^j(X, \mathcal{O}(-D)) \hookrightarrow H^j(Y, \mathcal{O}(-D'))$

$\therefore H^j(X, \mathcal{O}(-D)) = 0 \quad \forall j < \dim X.$ case 1 finished

case 2. For $E$ general.

Choose a log resolution of $(X, E)$. $Y \xrightarrow{h} X$

$h^*E$ has SNC support.

By Cor 4.1.4 (2) $\exists\; p \in \mathbb{N}^*$, $E' \in |ph^*A|$ eff. ($\therefore$ SNC support)

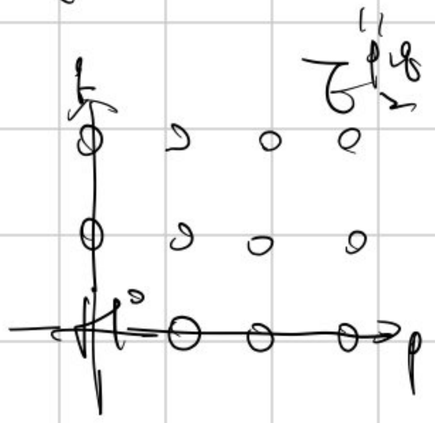
s.t. $ph^*A - E'$ ample

$mp \cdot h^*D \sim \underbrace{(p \cdot h^*A - E')}_{\text{ample}} + \underbrace{(E' + p \cdot h^*E)}_{\text{SNC support} \geq 0}$. Hence reduce to case 1

$\Rightarrow H^i(Y, \mathcal{O}_Y(K_Y + h^*D)) = 0 \quad \forall i \in \mathbb{N}^*$.

Leray $\Rightarrow H^p(X, R^q h_* \mathcal{O}_Y(K_Y + h^*D)) \Rightarrow H^{p+q}(Y, \mathcal{O}_Y(K_Y + h^*D))$.

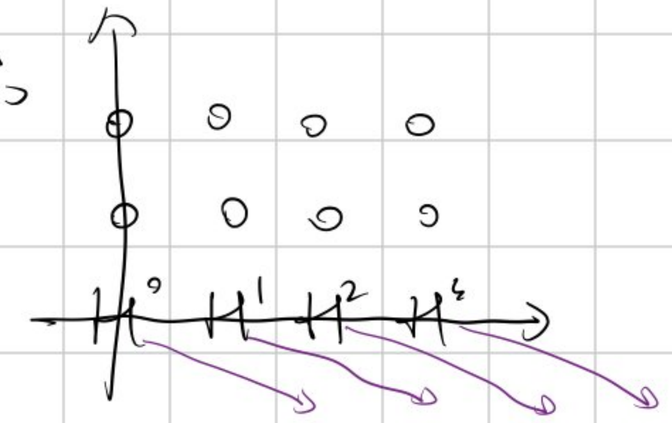
$E_\infty$-page:




Second page: projection formula & Cor 4.1.4(b)

$E_2^{p,q} = H^p(X, R^q h_* \mathcal{O}_Y(K_Y) \otimes \mathcal{O}_X(D))$

$= \begin{cases} 0 & q > 1 \\ H^p(X, \mathcal{O}_X(K_X + D)) & q = 0. \end{cases}$




$\Rightarrow H^p(X, \mathcal{O}_X(K_X + D)) = E_2^{p,0} = E_3^{p,0} = \cdots = E_\infty^{p,0} = \begin{cases} H^0(Y, \mathcal{O}_Y(K_Y + h^*D)) & p = 0 \\ 0 & p \neq 0 \end{cases}$ for $p \in \mathbb{N}$.

□

---

By Lefschetz hyperplane Thm and Hodge decomposition we have:

**Lemma 4.2.2.** *Let $X$ be a smooth complex projective variety of dimension $n$, and let $D \subseteq X$ be a smooth effective ample divisor. Denote by*

$$r_{p,q} : H^q(X, \Omega_X^p) \longrightarrow H^q(D, \Omega_D^p)$$

*the natural maps determined by restriction. Then*

$$r_{p,q} \text{ is } \begin{cases} \text{bijective} & \text{for } p+q \leq n-2, \\ \text{injective} & \text{for } p+q = n-1. \end{cases}$$

(Inspired by the idea of the pf of Kodaira's Vanishing, we hope:

**Theorem 4.2.3. (Nakano vanishing theorem).** *Let $X$ be a smooth complex projective variety of dimension $n$, and let $A$ be an ample divisor on $X$. Then*

$$H^q(X, \Omega_X^p \otimes \mathcal{O}_X(A)) = 0 \quad \text{for } p+q > n. \tag{4.1}$$

*Equivalently,*

$$H^s(X, \Omega_X^r \otimes \mathcal{O}_X(-A)) = 0 \quad \text{for } r+s < n. \tag{4.2}$$

(take $p = n$ $q > 0$ is just Kodaira Vanishing)

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"Analytically" define $\Omega^1_X \log D$ and $\Omega^p_X \log D$ for $D \subseteq X$ — nonsingular.

Suppose then that $X$ is a smooth variety, and $D \subset X$ is a non-singular divisor. As customary, we denote by $\Omega^1_X(\log D)$ the sheaf of one-forms on $X$ with logarithmic poles along $D$: if $z_1, \ldots, z_n$ are local analytic coordinates on $X$, with $D = (z_n = 0)$, then $\Omega^1_X(\log D)$ is locally generated by $dz_1, \ldots, dz_{n-1}, \frac{dz_n}{z_n}$. One puts

$$\Omega^p_X(\log D) = \Lambda^p\left(\Omega^1_X(\log D)\right).$$

Another way:

consider the conormal exact seq. $\quad i: D \hookrightarrow X$

$$0 \longrightarrow \mathcal{O}_D(-D) \longrightarrow i^*\Omega^1_X \longrightarrow \Omega^1_D \longrightarrow 0$$

**21.2.31. Theorem (conormal exact sequence for smooth varieties).** — *Suppose* $i: X \hookrightarrow Y$ *is a closed embedding of smooth varieties over a field* $k$, *with conormal sheaf* $\mathcal{N}^\vee_{X/Y}$. *Then* $\mathcal{N}^\vee_{X/Y}$ *is locally free, and the conormal exact sequence (Theorem 21.2.24(b)) is exact on the left:*

$$0 \longrightarrow \mathcal{N}^\vee_{X/Y} \xrightarrow{\delta} i^*\Omega_{Y/k} \longrightarrow \Omega_{X/k} \longrightarrow 0 \tag{21.2.31.1}$$

*is exact.*

$\mathcal{N}^\vee_{X/Y}$: $\mathcal{I}_D/\mathcal{I}_D^2$

$\rightsquigarrow \quad i^*(\Lambda^p\Omega^1_Y) = \Lambda^p i^*\Omega^1_Y \longrightarrow \Lambda^p\Omega^1_D$

adjoint $\quad \Omega^p_X \xrightarrow{\text{actually quotient}} i_*i^*\Omega^p_X \longrightarrow i_*\Omega^p_D$ (push forward of closed imm. is exact functor)

Define $\Omega^p_X(\log D) = \ker\left(\Omega^p_X \otimes \mathcal{O}_X(D) \longrightarrow i_*\Omega^p_D \otimes \mathcal{O}_X(D)\right)$.

A view of compatibility: locally the section in $\Omega^p_X \otimes \mathcal{O}_X(D)$ is of the form:

$\left(\sum_{i_1<\cdots<i_p} f_{i_1\cdots i_p}(z_1 \cdots z_n)\, dz_{i_1} \wedge \cdots \wedge dz_{i_p}\right) \otimes \frac{1}{z_n}$ By above morphism, the mapping is

$\longmapsto \left(\sum_{i_1<\cdots<i_p \le n+1} f_{i_1\cdots i_p}(z_1 \cdots z_{n+1}, 0)\, dz_{i_1} \wedge \cdots \wedge dz_{i_p}\right) \otimes \frac{1}{z_n}$

$f_{i_1\cdots i_p}(z_1 \cdots z_{n+1}, 0)$: a polynomial in $k[z_1 \cdots z_{n+1}]$

$\rightsquigarrow \Omega^p_X(\log D)$ locally of the form $\frac{1}{z_n}\left(z_n \sum_{i_1<\cdots<i_p} f_{i_1\cdots i_p}(z_1 \cdots z_n)\, dz_{i_1} \wedge \cdots \wedge dz_{i_p} + \sum_{i_n<\cdots<i_{p+1}\le n+1} f_{i_n\cdots i_{p+1}}(z)\, dz_{i_1} \wedge \cdots \wedge dz_{i_n}\right)$

"write" $z_n\left(\beta + \eta \wedge \frac{dz_n}{z_n}\right)$ $\quad \beta \in \Omega^p_X \quad \eta \in \Omega^{p-1}_X$

with "$\otimes \frac{1}{z_n}$", under different coordinate, it is compatible

$\Leftrightarrow$ gen. by $dz_1, dz_2, \ldots, \frac{dz_n}{z_n}$, $(p=1)$.

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**Lemma 4.2.4.** *Assume as above that $D \subseteq X$ is a smooth divisor.*

(i). *Viewing $\Omega^p_D$ as a sheaf on $X$ via extension by zero, one has exact sequences*

$$0 \longrightarrow \Omega^p_X \longrightarrow \Omega^p_X\big(\log D\big) \xrightarrow{res} \Omega^{p-1}_D \longrightarrow 0, \tag{4.3a}$$

$$0 \longrightarrow \Omega^p_X\big(\log D\big) \otimes \mathcal{O}_X(-D) \longrightarrow \Omega^p_X \longrightarrow \Omega^p_D \longrightarrow 0. \tag{4.3b}$$

(ii). *Let $\pi : Y \longrightarrow X$ be the $m$-fold cyclic covering branched along $D$ constructed in 4.1.6, and let $D' \subseteq Y$ be the divisor mapping isomorphically to $D$, so that $\pi^* D = mD'$. Then*

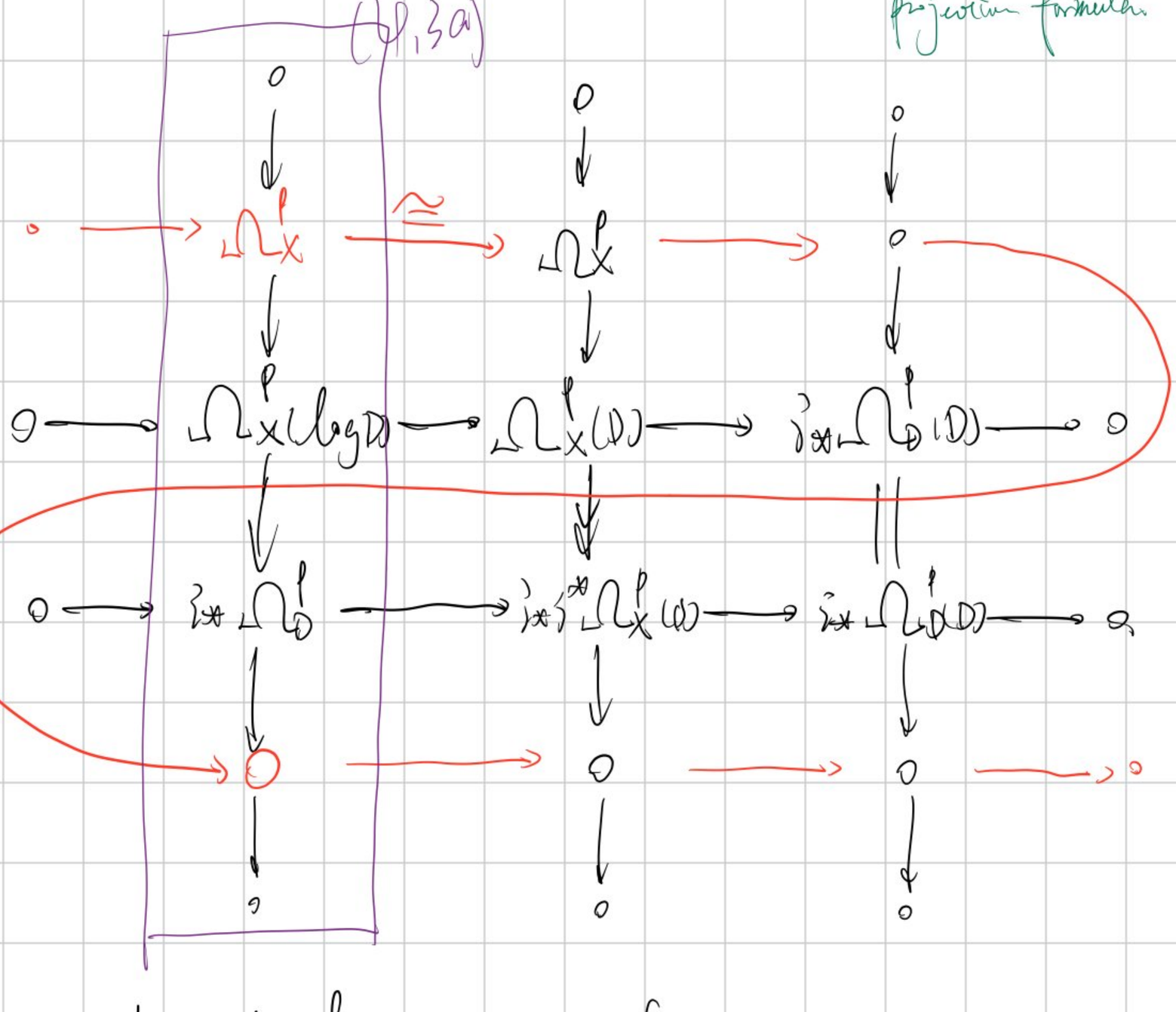
$$\pi^*\big(\,\Omega^p_X\big(\log D\big)\,\big) \;=\; \Omega^p_Y\big(\log D'\big).$$

in fact, (4.3b) can be the completely algebraic definition of $\Omega^p_X(\log D)$

Since $X$ smooth, $\Omega^p_X$ locally free

$\therefore 0 \to \Omega^p_X \to \Omega^p_X(D) \to i_* \mathcal{O}_D \otimes \Omega^p_X(D) \cong i_* i^* \Omega^p_X(D) \to 0$

(projection formula)




where the last row comes from:

$$0 \to \mathcal{O}_D(-D) \to i^* \Omega^1_X \to \Omega^1_D \to 0$$

$$\rightsquigarrow 0 \to \Omega^{p-1}_D \otimes \mathcal{O}_D(-D) \to i^* \Omega^p_X \to \Omega^p_D \to 0$$

$$\rightsquigarrow - \otimes \mathcal{O}_D(D) \rightsquigarrow i_* - \rightsquigarrow \checkmark$$

(rk=1)

reason: $0 \to L \to E \to F \to 0$

$0 \to L \otimes \Lambda^{p-1} F \to \Lambda^p E \to \Lambda^p F \to 0$,

$\phi(l \otimes v_1 \wedge \cdots \wedge v_p) = j(l) \wedge \tilde{v}_1 \wedge \cdots \wedge \tilde{v}_p$,

indep. from choice of lifting of $v_i \in F$ in $E$ (in essence $\Lambda^2 L = 0$)

Rmk: analytically "res" in (4.3a) sends "$\alpha \mapsto \gamma \wedge \frac{dg_n}{g_n}$" to $\overline{\gamma}$: restriction of $\gamma \in \Omega^{p-1}_X$ on $D$.

~~pf of (iii)~~ check affine locally:

$\mathrm{Spec}\, A[t]/(f^m - f) \longrightarrow \mathrm{Spec}\, A$ with $D|_{\mathrm{Spec}\,A} = (f)$.

$\Omega^1_X(\log D)(-D)$ over $\mathrm{Spec}\, A$ is $f\cdot\Omega^1_A + A\cdot df \subseteq \Omega^1_A \cong \Omega^1_A \oplus Ae \,/\, (df - fe)$

(construct $\Omega^1_A \oplus Ae \to \Omega^1_A$ mainly because $A$ UFD. $f$ prime element, $df$ can be extended to a $A$-basis of $\Omega^1_A$ free $A$-mod)

$\Omega^1_Y(\log D')(-D')$ over $\mathrm{Spec}\, B[t]/(t^m - f)$ is $\cong \Omega^1_B \oplus Be \,/\, (df - te)$

why can me'? reason.

$\Omega^1_B \oplus Be \,/\, (df - te) \cong B \otimes_A \Omega^1_A \oplus B\cdot dt \oplus B\cdot e \,/\, (df - mt^{m-1}dt,\ dt - te)$

$\cong B \otimes \Omega^1_A \oplus B\cdot e' \,/\, (df - f\cdot me')$

$B \otimes_A \Omega^1_A \oplus Be' \,/\, (df - mfe') \otimes_B B\cdot\frac{1}{t} \longrightarrow B \otimes_A \left( \Omega^1_A \oplus Ae \,/\, (df - fe) \otimes_A A\cdot\frac{1}{f} \right)$

by $A \to B$ faithfully flat.

$me' \longleftrightarrow e$

$(t\cdot\Omega^1_B + B\cdot dt) \otimes_B B\cdot\frac{1}{t} \longleftrightarrow B \otimes_A \left( (f\cdot\Omega^1_A + A\,df) \otimes_A A\cdot\frac{1}{f} \right)$

$dz_i \otimes \frac{1}{t} \qquad e' \otimes \frac{1}{t} \qquad dx_i \otimes \frac{1}{f} \qquad \frac{1}{m}\cdot e \otimes \frac{1}{f}$

$dz_i \qquad \frac{dt}{t} \qquad dx_i \qquad \frac{1}{m}\cdot\frac{df}{f}$

$\left( dt\otimes\frac{1}{t} = dt\otimes\frac{t^{m-1}}{f} = \frac{1}{m}mt^m dt\otimes\frac{1}{f} = \frac{1}{m}df\otimes\frac{1}{f} \right)$

$fe' \otimes \frac{1}{t} \qquad \frac{1}{m}fe\otimes\frac{1}{f}$

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**Theorem 4.2.3. (Nakano vanishing theorem).** *Let $X$ be a smooth complex projective variety of dimension $n$, and let $A$ be an ample divisor on $X$. Then*

$$H^q\big(X, \Omega_X^p \otimes \mathcal{O}_X(A)\big) = 0 \quad \text{for} \quad p+q > n. \tag{4.1}$$

*Equivalently,*

$$H^s\big(X, \Omega_X^r \otimes \mathcal{O}_X(-A)\big) = 0 \quad \text{for} \quad r+s < n. \tag{4.2}$$

Pf. prove by induction on $\dim X$. $\dim X = 1$, is Kodaira's Vanishing.
If the theorem holds for $\dim \le n-1$, for $\dim X = n$.

Choose $m \in \mathbb{N}^*$ and $D \in |mA|$ smooth. for $s+r<n$.

$$0 \to \Omega_X^r(-A) \to \Omega_X^r(\log D) \otimes \mathcal{O}_X(-A) \to \Omega_D^r(-A) \to 0.$$

$$\rightsquigarrow \cdots \to H^{s-1}\big(\Omega_D^r(-A)\big) \to H^s\big(\Omega_X^r(-A)\big) \to H^s\big(\Omega_X^r(\log D) \otimes \mathcal{O}_X(-A)\big) \to \cdots$$

By induction hypothesis $H^{s-1}\big(\Omega_D^r(-A)\big) = 0$.

Hope to show: $H^s\big(\Omega_X^r(\log D) \otimes \mathcal{O}_X(-A)\big) = 0$

$\pi: Y \to X$ cyclic covering along $D$. $D' \in |\pi^* A|$, $D' \cong D$.

By injective Lemma. it suffices to show

$$H^s\Big(Y, \pi^*\big(\Omega_X^r(\log D) \otimes \mathcal{O}_X(-A)\big)\Big) = 0 \qquad s+r<n.$$

Lem 2.4(ii) $\pi^*\big(\Omega_X^r(\log D)\big) \cong \Omega_Y^r(\log D')$. $\pi^*\mathcal{O}_X(-A) = \mathcal{O}_Y(-D')$.

By $0 \to \Omega_Y^r(\log D') \otimes \mathcal{O}_Y(-D') \to \Omega_Y^r \to \Omega_{D'}^r \to 0$.

$$\rightsquigarrow \cdots \to H^{s-1}\big(\Omega_Y^r\big) \xrightarrow{\text{big}} H^{s-1}\big(\Omega_{D'}^r\big) \to H^s\big(\Omega_Y^r(\log D') \otimes \mathcal{O}_Y(-D')\big)$$

$$\to H^s\big(\Omega_Y^r\big) \xrightarrow{\text{big on}} H^s\big(\Omega_Y^r\big) \to \cdots$$

By Lemma 4.2.2

$$\Rightarrow H^s\Big(\Omega_Y^r(\log D') \otimes \mathcal{O}_Y(-D')\Big) = 0. \qquad \blacksquare$$

**Example 4.3.7. (An extension of Theorem 4.3.1).** Let $D$ be a nef divisor on a smooth projective $n$-fold $X$. Assume that $\left(D^{n-k} \cdot H^k\right) > 0$ for some natural number $k$ and ample divisor $H$. Then

$$H^i\big(X, \mathcal{O}_X(K_X + D)\big) = 0 \quad \text{for} \quad i > k.$$

Rmk: when $k=0$. $D$ nef & $(D^n)>0$ i.e. $D$ big. it is just Kawamata's Vanishing.

Pf. prove by induction on $\dim X = n$. ($n=1$, trivial) if holds for $\dim \le n-1$.

WLOG $H$ very ample & effective & smooth. irreducible.

$$0 \to \mathcal{O}_X(K_X+D) \to \mathcal{O}_X(K_X+D+H) \to \mathcal{O}_H(K_X+D+H)|_H) \to 0.$$

$$\mathcal{O}_H(K_H + D|_H)$$

$$\rightsquigarrow H^{i-1}\big(H, \mathcal{O}_H(K_H+D|_H)\big) \to H^i\big(X, \mathcal{O}_X(K_X+D)\big) \to H^i\big(X, \mathcal{O}_X(K_X+D+H)\big) \quad (i>k)$$

$(D^{n-k}\cdot H^k) = \big(D|_H^{n-k} \cdot \underbrace{H|_H^{k-1}}_{\text{ample on } H}\big) > 0$. $D|_H$ nef on $H$.

Since $i-1 > k-1$. By induction hypothesis $H^{i-1}\big(H, \mathcal{O}_H(K_H+D|_H)\big) = 0$

$D+H$ ample. by Kodaira's Vanishing $H^i\big(X, \mathcal{O}_X(K_X+D+H)\big) = 0$.

$\therefore$ $H^i\big(X, \mathcal{O}_X(K_X+D)\big) = 0$. ▨

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